



First Semester 2026-27 Course Handout Part II

Date: 31 July 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number	: ME F428/ ME U428
Course Title	: SMART MATERIALS
Instructor-in-charge	: Dr. Hemant Sharma (Chamber: 2219-J)
Team of Instructors	: Not applicable

1. Course description: The course introduces smart materials and smart structures through their physical mechanisms, characteristic responses and engineering use. Major topics include piezoelectric ceramics and polymers; one-way, two-way and magnetic shape-memory alloys; electro- and magnetorheological fluids; magnetostrictive and electrostrictive materials; and fibre-optic and FBG sensors. Basic modelling is embedded within the relevant material modules using physically interpretable equations and response curves. Applications include sensing, actuation, vibration control, damping, structural-health monitoring, energy harvesting and intelligent devices.

2. Scope and Objective: This undergraduate course develops a foundational understanding of stimulus-responsive materials and their use in smart structures. Emphasis is placed on physical mechanisms, observable behaviour, characteristic curves and straightforward engineering comparisons rather than advanced constitutive modelling. Guided demonstrations, basic experiments, case studies, paper reviews, assignments and project work will connect theory with practice. These activities are intended to develop scientific curiosity and encourage interested students to pursue further project- and research-oriented work in smart materials and structures.

3. Course learning outcomes:

CO1: Explain the terminology, operating principles, physical mechanisms and characteristic behaviour of major smart-material families and smart structural systems.

CO2: Interpret basic characteristic curves, experimental observations and performance data to describe the sensing and actuation behaviour of smart materials.

CO3: Compare major smart-material families and identify suitable materials for simple engineering applications based on their capabilities, limitations and operating requirements.

CO4: Investigate smart-material behaviour through guided demonstrations, basic experiments, case studies, paper reviews or project work, and communicate the resulting observations and findings.

4. Textbooks and references:

TB1. H. Abramovich, *Intelligent Materials and Structures*, 2nd Edition, De Gruyter, 2021.

TB2. I. Chopra and J. Sirohi, *Smart Structures Theory*, Cambridge University Press, 2014.

RB1. M. V. Gandhi and B. S. Thompson, *Smart Materials and Structures*, Chapman & Hall, 1992.

RB2. Class notes, demonstrations, experiments and specified self-study resources announced during the course.

5. Course Format & Weekly Workflow

Indicative 14-week schedule: three scheduled sessions per week. Module boundaries may extend into adjacent weeks depending on demonstrations, quizzes and class progress.

Week	Module	Topics	Learning Objectives / Indicative Coverage	Book (Chapter)	CO(s)
1	M1	Introduction and historical perspective	Introduction to smart materials and smart structures; historical development of the field; basic terminology, classification and fundamental characteristics of adaptive, intelligent and multifunctional systems.	TB2 (1); RB1 (1-2)	CO1
2	M2	Smart and intelligent structural systems and applications	Overview of sensing, actuation and adaptive-system concepts; passive, active and semi-active structures; integration of smart materials within engineering systems; representative present and emerging applications.	TB1 (1); RB1 (3)	CO1; CO3
3	M3	Review of major smart-material families	General introduction and comparison of the principal smart-material families based on activating stimulus, physical response, engineering function, operating characteristics and application potential.	TB1 (1); RB2	CO1; CO3
4	M4	Shape-memory alloys	Introduction to shape-memory alloys; physical principles and solid-state phase-transformation behaviour.	TB1 (4); RB2	CO1
5	M4	Shape-memory alloys	Thermally and mechanically induced transformations; transformation temperatures; one-way and two-way shape-memory effects; superelasticity; interpretation of characteristic response curves and guided demonstration observations.	TB1 (4); RB2	CO1; CO2; CO4
6	M4	Shape-memory alloys	Simplified response models; actuation methods; magnetic shape-memory alloys; representative engineering applications, selection considerations and practical limitations.	TB1 (4); RB2	CO2; CO3
7	M5	Electro- and magnetorheological fluids	Composition, operating principles and field-dependent behaviour of ER and MR fluids; simplified rheological representation; common operating configurations; semi-active devices; guided field-response demonstration; engineering applications and practical performance considerations.	TB1 (5); RB2	CO1; CO2; CO3; CO4
8	M6	Laminated composites and bonded smart structures	Essential concepts of composite laminates required for understanding structures containing bonded or embedded smart-material layers; basic material behaviour, stress-strain relations and selected applications in active beams, plates and structural components.	TB1 (2); RB2	CO1; CO2; CO3
9	M7	Piezoelectric materials	Fundamentals and classification of piezoelectric materials, including ceramics and polymers; direct and converse piezoelectric effects.	TB1 (3); RB2	CO1
10	M7	Piezoelectric materials	Essential material properties; simplified electromechanical constitutive behaviour; basic experiment or demonstration involving sensing and actuation.	TB1 (3); RB2	CO2; CO4
11	M7	Piezoelectric materials	Structural integration and dynamic response; piezoelectric vibration sensing and control; energy-harvesting principles; representative application case studies.	TB1 (3); RB2	CO2; CO4
12	M8	Fibre-optic and FBG sensors	Basic optical principles relevant to fibre sensing; construction and characteristics of optical fibres and fibre Bragg gratings; strain and temperature measurement; interrogation principles, multiplexing, guided sensor demonstration and structural-health-monitoring applications.	TB1 (9); RB2	CO1; CO2; CO4

Week	Module	Topics	Learning Objectives / Indicative Coverage	Book (Chapter)	CO(s)
13	M9	Magnetostrictive and electrostrictive materials	Physical mechanisms and field-induced responses of magnetostrictive and electrostrictive materials; characteristic behaviour; sensing and actuation potential; representative applications and limitations.	TB1 (6); RB2	CO1; CO2; CO3
14	M10	Applications of smart materials in integrated systems	Integration of smart materials into engineering devices and structures; representative applications across major industrial sectors; comparison and selection of material technologies; introductory considerations of dynamics, control, reliability and safety; project presentations or paper-review discussions.	TB2 (8); RB2	CO3; CO4
	Total	40 lectures			

- Regular demonstrations and experiments will be conducted to help students understand the actual behaviour and physical performance of different smart materials.

6. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
Quizzes 1, 2 and 3	10%	20 min each	Two announced; one unannounced	In-class, closed book; best two scores count, carrying 5% each.
Mid-Semester Examination	30%	90 min	As per the timetable	Closed book.
Demonstrations, Experiments, Project and Assignments	20%	Continuous	TBA / continuous	In-class and/or take-home; open book. Activity-specific instructions will be announced.
Comprehensive Examination	40%	3 hours	As per the timetable	Closed book/open book as announced.
Total	100%			

Criterion for NC: Scoring less than 10% of the course top score or completely missing a major evaluation component (Mid-Semester Examination or Comprehensive Examination) will attract an NC grade, subject to prevailing Institute regulations.

- Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
- Notices:** If any, will be communicated through Nalanda.
- Chamber consultation hours:** Will be announced in the class.
- Makeup Policy:** Make-up will be permitted only in genuine cases with prior permission and in accordance with Institute rules.

Dr. Hemant Sharma
Instructor-in-charge
Course No. ME F428